01 · The approach

One engine, two markets: keep what transfers, re-measure the rest, test both ways.

One engine. Two markets.

WEEK 1 · DEVELOPED MARKET

#3

of 54 teams

private D^2 0.13864

2.4 million policy rows

WEEK 2 · EMERGING MARKET

#2

private leaderboard

public D^2 0.16458

2 years of data · 76% new business

1. **Keep what transfers**
   ranking of risk from the developed market
2. **Re-measure what does not**
   level and mix from the emerging book
3. **Test every step both ways**
   forward and reverse; neither may get worse

AI ALONGSIDE ME, AT EVERY STEP

- 🔎 **explore the data**
- 🧪 **code + 125 tests**
- ⚙️ **experiments**
- ✅ **verify its own work**

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02 · What changed

The new market differs by peril and by customer mix, not by one level factor.

+34%

motor claim frequency

−42%

motor mean claim cost

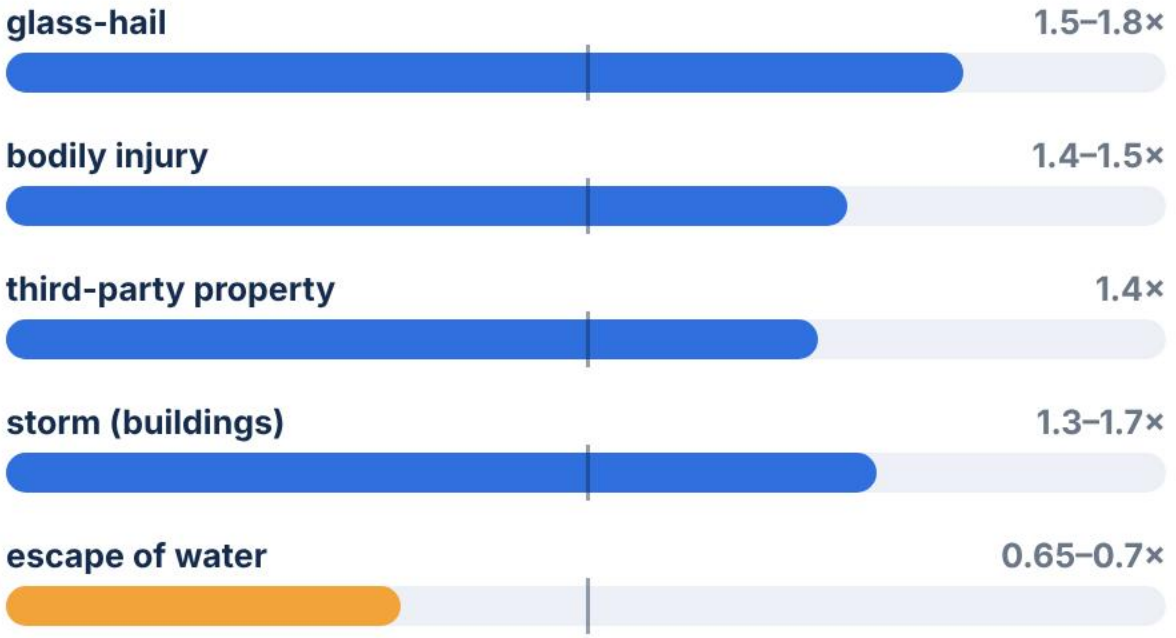


76%

of the 2025 test book is new business
no claim history

Claim frequency, emerging ÷ developed

ranges measured in both years · 1.0× = same as developed



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03 · The method

The developed market decides who is risky; the emerging book decides how much.

Developed model

2.4M rows · variables both markets share

+ Claim history

last year: policy · address · customer

Ensemble

5 trees · 7 nets · 4 Tweedie

Level from emerging book

counts × winsorised severity · credibility-weighted

Gender factor

one flat factor per line

The ensemble: 16 models, averaged in groups

5 tree models

7 neural nets

4 Tweedie GBMs

The emerging data decides how much. The developed data decides who.

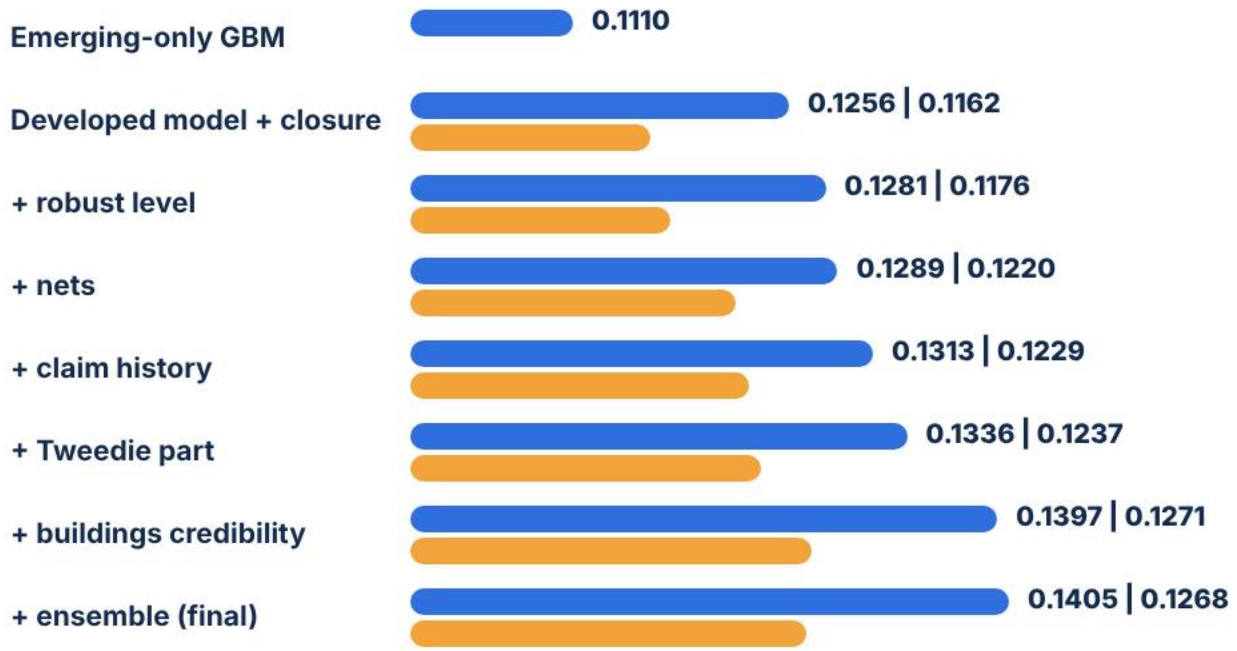
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04 · How we trust it

Every change is kept only if it improves the score in both time directions.

Pooled D^2 on the emerging book, every step tested both ways

forward (calibrate 2023 → score 2024) reverse (2024 → 2023) · axis starts at 0.10



30+ ideas rejected

- ✕ Pooled training with a market flag
- ✕ GLM in the blend
- ✕ Quantile-input net on its own
- ✕ Clipping high predictions
- ✕ Cross-line covariates
- ✕ Shared peril multipliers
- ✕ Domain-shift weights
- ✕ Slope calibration
- ✕ Multi-seed level refits
- ✕ Emerging-only models
- ✕ Per-line severity constants
- ✕ Claim history split by line
- ✕ ...and 18 more

SIMULATION

known truth: robust level beats plain closure

CEILING

≤ +0.005

D^2 from a perfect level

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05 · AI in the workflow

AI does the legwork; verification and decisions stay with me, and are recorded.

Parallel helper sessions, refitting while I was away

HELPER 1 3-seed GBM + Tweedie	HELPER 2 3-seed TabM, history	HELPER 3 TabM nets, history
HELPER 4 TabM-PLE nets, history	HELPER 5 direct Tweedie GBM	HELPER 6 MIX construction
HELPER 7 GBM settings screen	HELPER 8 market-normalised features	HELPER 9 premium-free model
HELPER 10 entity embeddings	HELPER 11 recency & domain shift	HELPER 12 XGBoost & CatBoost

1 Unsupported statements

Three claims in the first data review had no evidence: replaced by measured values before commit.

2 A bias that was a bug

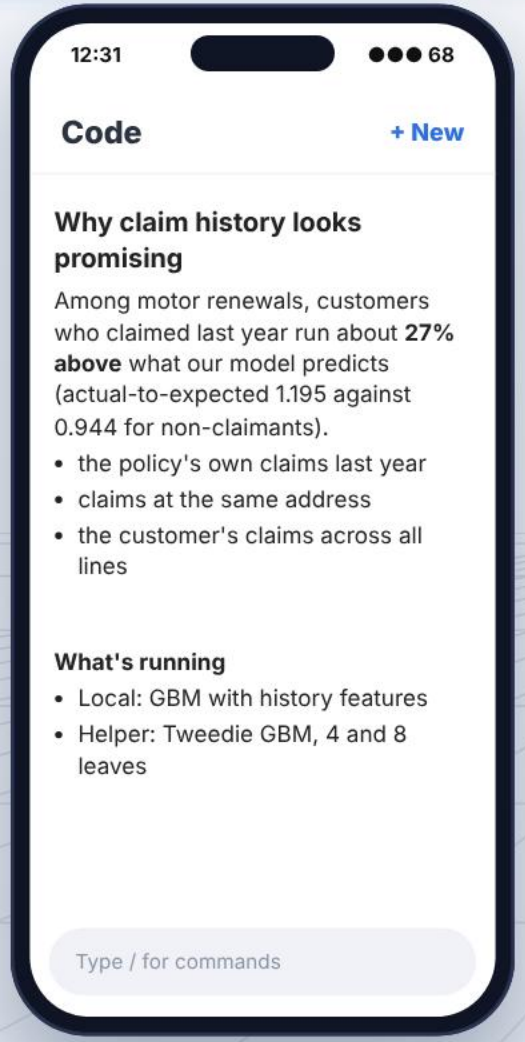
The simulation showed every estimator biased low; the plain closure had it too: the test harness was wrong, conclusion withdrawn.

3 Already rejected

A net we built had failed in Part 1 on the big folds; the public score agreed. We now search the record first.

06 · A new way to work

Example: part of week 2 was done from my phone, by voice, on a bus.



Modelled on the move.

Part of week 2, on a bus · phone only · voice only

AI-assisted actuarial work, away from the desk.

10+
parallel helper sessions

12
Part 2 submissions

125
automated tests

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07 · Guardrails and uncertainty

Guardrails and stated limits: what could go wrong, and how we would notice.

🚨

Leak alarm

D^2 above 0.35 stops a run

🔁

Reproducible

re-run matches to 1e-6

📋

Model card

monitoring triggers

🧾

84 decisions

each with its evidence

Main stated uncertainty: the buildings level



Limits we state

⚖️ Gender is a rating factor: flagged for regulatory review; a gender-free file is the alternative

📉 Two emerging years, 76% new business: the level, not the ranking, is the risk

🧪 Public scores within 0.0002: we select by local validation

Guardrails and stated limits: what could go wrong, and how we would notice.

Kohei Kudo · IAA AI-Assisted Hackathon 2026

Built with an AI teammate, from a bus seat.

Week 1 · #3 of 54 | Week 2 · #2

github.com/wideflat/iaa-hackathon2026

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